

Prepared By	09-Jun-2012	Mr.Chatchawan S.	 RENEWCOILS ENGINEER & SUPPLY CO.,LTD. Renew coils Engineering & Supply Co., Ltd.	
Checked By	09-Jun-2012	Mr.Chatchawan S.		
Approved By	09-Jun-2012	Mr.Amnaury W.		
Certified By	09-Jun-2012	Mr.Amnaury W.		
Rev.	Date	Checked	Approved	Description
0	09-Jun-2012	Mr.Chatchawan S.	Mr.Amnaury W.	Issue for Approved

Renew Coils Engineering & Supply Co., Ltd.

Doc. No. : RNC-WPS-013-2012

Addendum

**WELDING PROCEDURE SPECIFICATION (WPS)
RNC-WPS-FCAW-001**



WELDING PROCEDURE SPECIFICATION (WPS)

WELDING PROCEDURE SPECIFICATION (WPS) ☒ **Yes**
 PREQUALIFIED ☒ **X** QUALIFIED BY TESTING ☐
 or PROCEDURE QUALIFICATION RECORD (PQR)

Identification # RNC-WPS-FCAW-001

Company Name Renew coils and Engineering & Supply co.,lt Authorized by QA/QC By Renee coil and Engineering
 Welding Process (es) FCAW Type-Manual ☐ Date 09 Jun'2012 Semi-Automatic ☒
 Supporting PQR No.(s) RNC-PQR-FCAW-001 Machine ☐ Automatic ☐

JOINT DESIGN USED

Type : Butt Weld
 Single ☒ Double Weld ☐
 Backing : Yes ☐ No ☒
 Backing Material : NONE
 Root Opening 3 Root Face Dimension
 Groove Angle 60^{0±5} Radius (J-U) NONE
 Back Gouging ☒ Yes ☐ No ☐ Method Electrode Gouging,
Air Gouging Carbon or Grinding

BASE METALS

material Spec. SS 400 or Strength Equivalence
 Type or Grade Group 1 Carbon
 Thickness : Groove 3.0 mm - Unlimited Fillet N/A
 Diameter (Pipe)

FILLER METALS

AWS Specification AWS A5.20
 AWS Classification E71T-1
 Brand Name KobelcoDW-100

SHIELDING

Flux NONE CO₂
 Composition 99.99%
 Electrode-Flux (Class) E71T-1 Flow Rate 15-20 L/min
 Gas Cup Size # 6-12

PREHEAT

Preheat Temp., Min 30 C°
 Interpass Temp., Min 100C° Max 250 C°

POSITION

Position of Groove : All Fillet : None
 Vertical Progression : Up ☒ Down ☐

ELECTRICAL CHARACTERISTICS

Transfer Mode (GMAW) Short-Circuiting ☐
 Globular ☐ Spray ☐
 Current : AC ☐ DCEP ☒ DCEN ☐ Pulsed ☐
 Other
 Tungsten Electrode (GTAW)
 Size : NONE
 Type : NONE

TECHNIQUE

Stringer or Weave Bead : Stringer and Weave Bead
 Multi-pass or Single Pass (per side) Multi-pass
 Number of Electrodes 1
 Electrode Spacing
 Longitudinal N/A
 Lateral N/A
 Angle N/A
 Contact Tube to Work Distance 10 - 15 mm
 Peening N/A
 Interpass Cleaning : Grinding or Wire Brush

POSTWELD HEAT TREATMENT

Temp. N/A
 Time N/A

WELDING PROCEDURE

Pass or Weld Layer(s)	Process	Filler Metals		Current		Volts	Travel Speed cm/min	Joint Details
		Class	Diam.	Type & Polarity	Amps or Wire Feed Speed			
Root pass	FCAW	E71T-1	1.2	DC EP	130-170	20-24	13-18	
Hot pass	FCAW	E71T-1	1.2	DC EP	200-220	27-30	20-28	
Filling pass	FCAW	E71T-1	1.2	DC EP	200-225	27-30	30-35	
cap	FCAW	E71T-1	1.2	DC EP	190-200	27-31	30-35	

PREPARED BY:

REVIEWED BY:

APPROVED BY:

DATE:

DATE:

DATE:

WELDING PROCEDURE SPECIFICATION (WPS) Yes ☐
 PREQUALIFIED _____ QUALIFIED BY TESTING _____
 or PROCEDURE QUALIFICATION RECORD (PQR) Yes ☒
 (ACCORDANCE WITH THE REQUIREMENT OF SEC'T 4 AWS D1.1 2008)

Company Name		RENEWCOILS ENGINEERING & SUPPLY CO.,LTD.		Identification #		RNC-WPS-FCAW-001	
Welding Process(es)		FLUX CORED ARC WELDING (FCAW)		Revision		1	
Supporting PQR No.(s)		RNC-PQR-FCAW-001		Date		9 Jun 12	
				By		RNC	
				QA/QC		Date	
						9 Jun 12	
				Type		☐ Manual ☐ Machine	
						☒ Semi-Automatic ☐ Automatic	

JOINT DESIGN USED

Type : Single V-Groove Butt Weld

☒ Single ☐ Double weld

Backing : ☐ Yes ☒ No

Backing Material : None

Root Opening : 3.0 mm. Root Face Dimension : 3.0 mm.

Groove Angle : 60° ±5 Radius (J-U) : -

Back Gouging ☒ Yes ☐ No

Method : Grinding

BASE METALS			
Material Spec.	Carbon Steel		
Type or Grade	SS 400		
Thickness :	Groove	25 mm.	Fillet : -
Diameter (Pipe)	-		

FILLER METALS	
AWS Specification	A 5.20
AWS Classification	E71T-1

SHIELDING			
Flux	NONE	Gas	CO ₂
		Composition	99.99 %
Electrode-Flux(Class)	-	Flow Rate	18 L/min.
		Gas Cup Size	N/A

PREHEAT			
Preheat Temp.:	Ambient		
Interpass Temp., Min	100°C	Max	210°C

POSITION
Position of Groove 2G Fillet F,H
Vertical Progression Up Down

ELECTRICAL CHARACTERISTICS

Transfer Mode (GMAW)	☐ Short-Circuiting	☐ Spray/Pulse
	☐ Globular	☐ Pulsed
Current	☐ AC ☐ DCEP	☒ DCEN
Other	-	
Tungsten Electrode (GTAW)		
Size	None	
Type	None	

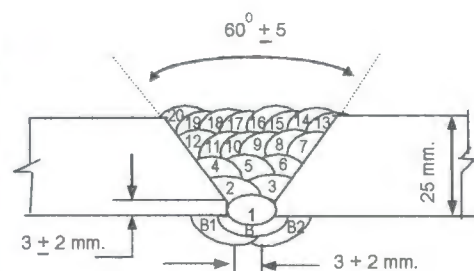
TECHNIQUE	
Stringer or Weave Bead :	Stringer
Multi-pass or Single Pass (per side)	Multipass
Number of Electrodes	Single
Electrode Spacing	Longitudinal N/A
	Lateral N/A
	Angle N/A

Contact End to Work Distance	15 mm.
Peening	None
Interpass Cleaning :	Slags remove by Grinding / Wire Brush

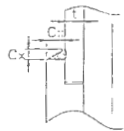
POSTWELD HEAT TREATMENT	
Temperature	N/A
Time	N/A

WELDING PROCEDURE

Pass or Weld Layer(s)	Process	Filler Metals		Current		Volts	Travel Speed (cm / min)	Heat Input (kJ/cm)
		Class	Dia. (mm.)	Type & Polarity	Amperage or Wire speed			
1	FCAW	E71T-1	1.2	DCEP	170	24	18	13.60
2	FCAW	E71T-1	1.2	DCEP	170	24	32	7.65
3	FCAW	E71T-1	1.2	DCEP	210	27	35	9.72
4	FCAW	E71T-1	1.2	DCEP	210	30	30	12.60
5	FCAW	E71T-1	1.2	DCEP	200	30	32	11.25
6	FCAW	E71T-1	1.2	DCEP	210	30	34	11.12
7	FCAW	E71T-1	1.2	DCEP	210	30	30	12.60
8	FCAW	E71T-1	1.2	DCEP	200	30	35	10.29
9	FCAW	E71T-1	1.2	DCEP	210	30	35	10.80
10	FCAW	E71T-1	1.2	DCEP	200	30	35	10.29
11	FCAW	E71T-1	1.2	DCEP	200	31	30	12.40
12	FCAW	E71T-1	1.2	DCEP	200	30	35	10.29
13	FCAW	E71T-1	1.2	DCEP	210	30	35	10.80
14	FCAW	E71T-1	1.2	DCEP	210	30	35	10.80
15	FCAW	E71T-1	1.2	DCEP	210	30	35	10.80
16	FCAW	E71T-1	1.2	DCEP	210	30	35	10.80
17	FCAW	E71T-1	1.2	DCEP	210	30	35	10.80
18	FCAW	E71T-1	1.2	DCEP	210	30	35	10.80
19	FCAW	E71T-1	1.2	DCEP	210	29	35	10.44
20	FCAW	E71T-1	1.2	DCEP	210	29	35	10.44
Back Weld	FCAW	E71T-1	1.2	DCEP	200	27	32	10.13
B1	FCAW	E71T-1	1.2	DCEP	200	28	30	10.20
B2	FCAW	E71T-1	1.2	DCEP	200	28	35	9.60

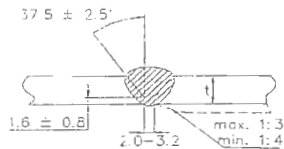


ATTACHED FIGURE JOINT SKETCH



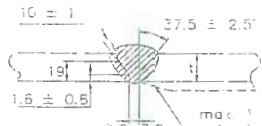
C_x (min.) = $1.25t$ but not less than 3.2 mm.

JOINT No. A1



Misalignment (max.) = 1.5 mm

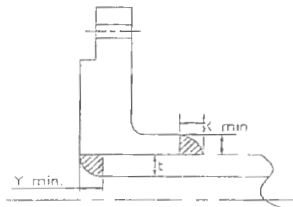
JOINT No. B1 & B2



Approx. 15 mm.

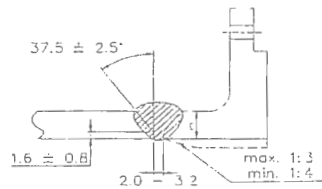
X min. = The lesser of $1.4t$ or thickness of the hub.

JOINT No. A2



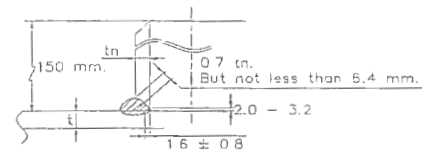
X min. = The lesser of $1.4t$ or thickness of the hub.
 Y min. = The lesser of t or 6.4 mm.

JOINT No. A3



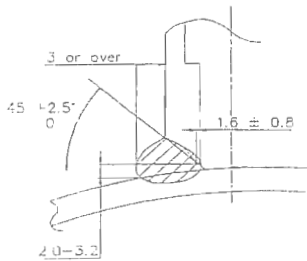
Misalignment (max.) = 1.5 mm

JOINT No. B3

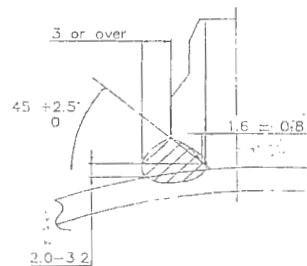


Misalignment (max.) = The lesser of 3.2 mm. or 0.5tn.

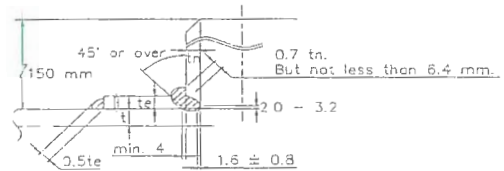
JOINT No. B4



JOINT No. C1

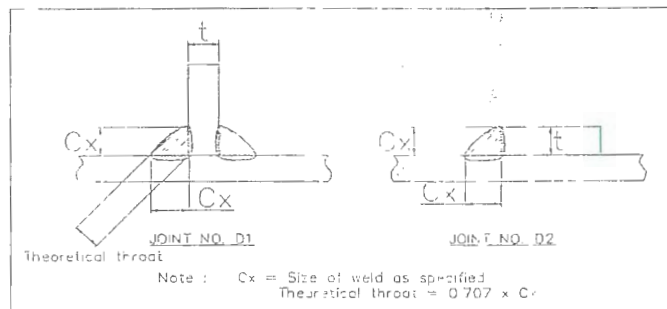


JOINT No. C2



Misalignment (max.) = The lesser of 3.2 mm. or 0.5tn.

JOINT No. B5





PROCEDURE QUALIFICATION RECORD (PQR)

NDT and Mechanical Test Result

TENSILE TEST (Report No : TS-12-06-108)						
Specimen No.	Width mm.	Thickness mm.	Area	Ultimate Load (KN)	Ultimate Strength (N/mm. ²)	Character of Failure and Location
2369/12TR1	20.37	24.84	505.991	244.381	482.976	OUT OF WELD
2369/12TR2	20.52	24.85	509.922	244.381	479.253	OUT OF WELD

GUIDE BEND TEST (Report No : BD-12-06-106)			
Specimen No	Type of Bend	Test Result	Location of Indication
2369/12BS1	Transverse Side Bend	No Open Discontinuity	N/A
2369/12BS2	Transverse Side Bend	No Open Discontinuity	N/A
2369/12BS3	Transverse Side Bend	No Open Discontinuity	N/A
2369/12BS4	Transverse Side Bend	Open Discontinuity 1.10 mm	WELD ZONE

Visual Inspection, PQR Report No:

PQR-007

Appearance: Acceptable
Undercut/Slags: No Indication
Surface Porosity: No Indication
Test Date: 15 Jun 12
Witness By: Mr.Natthawut M.
Organization: PAE Technical Service Co.,Ltd.

Radiographic Examination

RT Report No: PQR-007 Result: Accepted
Organization: PAE Technical Service Co.,Ltd.

Other Tests

1) Vicker Hardness Test

Report No: HN-12-06-083

2) Macrostructure Examination

Report No: MA-12-06-105

3) Fillet Weld Test

Report No: N.A.

4) Impact Test

Report No: IM-12-06-134

Welder's Name: Mr.Anan Y.

Welder No: -

Mechanical Test Conducted By: LPN Plate Mill Public Company Limited

Test Conducted By: Mr.Keerati Sa-ngoen

Report No: -

We, the undersigned, certify the statements in this record are correct and that the test welds were prepared, welded, and Tested in conformance with the requirements of Clause Section 4 of AWS D1.1 Structural Welding Code-Steel.

Manufacturer by
Rayong Maintenance & Contracting Co.,Ltd.

Contractor by
PAE Technical Service Co.,Ltd.

Signature: _____
Name: _____
Date: _____



Signature: _____
Name: Mr. Chabemkiet R.
Date: 20 Jun 2012



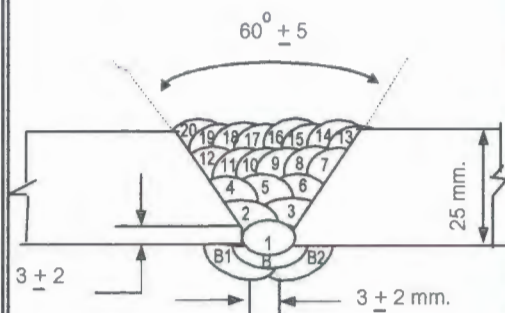
WELDER PROCEDURE QUALIFICATION RECORD

Page No. 1 of 3

Client : Renew Coils Engineering & Supply Co.,Ltd.		Project Name : PQR		Location : RNC Work Shop		Report No. PQR-007	
		Customer Order No. : N/A		WPS No. : RNC-WPS-FCAW-001		Date : 15-Jun-12	

Welder No.	Welder Name	Welding Process	Test Position	Applicable Standard	Material		Electrode Specification	Results	
					Spec	Size		Visual	RT
-	MR.ANAN Y.	FCAW	2G	AWS D1.1	SS400	350x150x25	E71T-1	ACCEPTED	ACCEPTED

Joint Details



Layer (S)	Process	Filler Metal		Current		Volt	Travel Speed cm/min	Interpas Temp °C	Heat Input (KJ / cm)	Shielding		Backing	
		Class	Dia. (mm.)	Type Polarity	Amp					GAS	Flow Rate (l/min)	GAS	Flow Rate (l/min)
1	FCAW	E71T-1	1.2	DCEP	170	24	18	100	13.60	CO ₂	18	-	-
2	FCAW	E71T-1	1.2	DCEP	170	24	32	113	7.65	CO ₂	18	-	-
3	FCAW	E71T-1	1.2	DCEP	210	27	35	120	9.72	CO ₂	18	-	-
4	FCAW	E71T-1	1.2	DCEP	210	30	30	124	12.60	CO ₂	18	-	-
5	FCAW	E71T-1	1.2	DCEP	200	30	32	140	11.25	CO ₂	18	-	-
6	FCAW	E71T-1	1.2	DCEP	210	30	34	145	11.12	CO ₂	18	-	-
7	FCAW	E71T-1	1.2	DCEP	210	30	30	154	12.60	CO ₂	18	-	-
8	FCAW	E71T-1	1.2	DCEP	200	30	35	161	10.29	CO ₂	18	-	-
9	FCAW	E71T-1	1.2	DCEP	210	30	35	184	10.80	CO ₂	18	-	-
10	FCAW	E71T-1	1.2	DCEP	200	30	35	191	10.29	CO ₂	18	-	-

PAE Inspection By

Witness By

Approved By

SIGNATURE :

NAME :

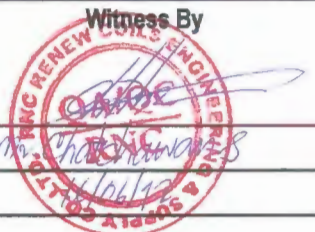
DATE :



SIGNATURE :

NAME :

DATE :



SIGNATURE :

NAME :

DATE :

[Signature]

09/06/12



PAE TECHNICAL SERVICE CO.,LTD

NDT, TESTING, INSPECTION & ENGINEERING CONSULTANT

69 SOI.OONNUT 64 (SUKSAMARN) SRINAKARIN RD, SUANLUANG,BANGKOK 10250

Tel : 02) 322-0222, Fax : 02)721-2577, RAYONG (038) 607-402-3

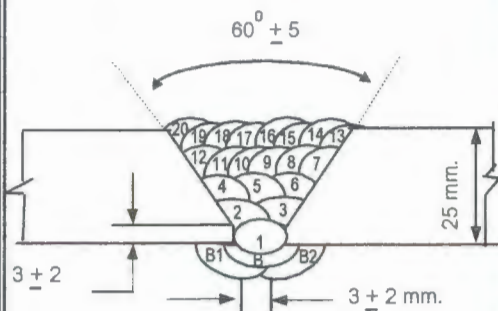
WELDER PROCEDURE QUALIFICATION RECORD

Page No. 2 of 3

Client :	Project Name : PQR	Location : RNC Work Shop	Report No. PQR-007
Renew Coils Engineering & Supply Co.,Ltd.	Customer Order No. : N/A	WPS No. : RNC-WPS-FCAW-001	Date : 15-Jun-12

Welder No.	Welder Name	Welding Process	Test Position	Applicable Standard	Material		Electrode Specification	Results	
					Spec	Size		Visual	RT
-	MR.ANAN Y.	FCAW	2G	AWS D1.1	SS400	350x150x25	E71T-1	ACCEPTED	ACCEPTED

Joint Details



Layer (S)	Process	Filler Metal		Current		Volt	Travel Speed cm/min	Interpas Temp °C	Heat Input (KJ / cm)	Shielding		Backing	
		Class	Dia. (mm.)	Type Polarity	Amp					GAS	Flow Rate (l/min)	GAS	Flow Rate (l/min)
11	FCAW	E71T-1	1.2	DCEP	200	31	30	195	12.40	CO ₂	18	-	-
12	FCAW	E71T-1	1.2	DCEP	200	30	35	192	10.29	CO ₂	18	-	-
13	FCAW	E71T-1	1.2	DCEP	210	30	35	183	10.80	CO ₂	18	-	-
14	FCAW	E71T-1	1.2	DCEP	210	30	35	185	10.80	CO ₂	18	-	-
15	FCAW	E71T-1	1.2	DCEP	210	30	35	187	10.80	CO ₂	18	-	-
16	FCAW	E71T-1	1.2	DCEP	210	30	35	197	10.80	CO ₂	18	-	-
17	FCAW	E71T-1	1.2	DCEP	210	30	35	199	10.80	CO ₂	18	-	-
18	FCAW	E71T-1	1.2	DCEP	210	30	35	193	10.80	CO ₂	18	-	-
19	FCAW	E71T-1	1.2	DCEP	210	29	35	210	10.44	CO ₂	18	-	-
20	FCAW	E71T-1	1.2	DCEP	210	29	35	192	10.44	CO ₂	18	-	-

PAE Inspection By

Witness By

Approved By

SIGNATURE :

NAME :

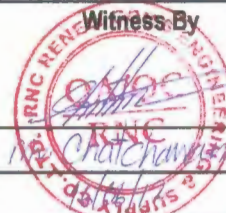
DATE :



SIGNATURE :

NAME :

DATE :



SIGNATURE :

NAME :

DATE :

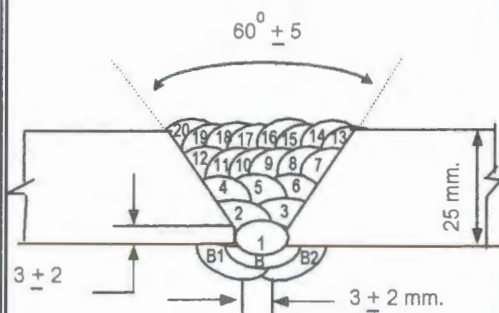
[Signature]
09/06/12

WELDER PROCEDURE QUALIFICATION RECORD

Page No. 3 of 3

Client : Renew Coils Engineering & Supply Co.,Ltd.		Project Name : PQR		Location : RNC Work Shop		Report No. PQR-007			
		Customer Order No. : N/A		WPS No. : RNC-WPS-FCAW-001		Date : 15-Jun-12			
Welder No.	Welder Name	Welding Process	Test Position	Applicable Standard	Material		Electrode Specification	Results	
					Spec	Size		Visual	RT
-	MR.ANAN Y.	FCAW	2G	AWS D1.1	SS400	350x150x25	E71T-1	ACCEPTED	ACCEPTED

Joint Details



Layer (S)	Process	Filler Metal		Current		Volt	Travel Speed cm/min	Interpas Temp °C	Heat Input (KJ / cm)	Shielding		Backing	
		Class	Dia. (mm.)	Type Polarity	Amp					GAS	Flow Rate (l/min)	GAS	Flow Rate (l/min)
B	FCAW	E71T-1	1.2	DCEP	200	27	32	101	10.13	CO ₂	18	-	-
B1	FCAW	E71T-1	1.2	DCEP	200	28	30	163	11.20	CO ₂	18	-	-
B2	FCAW	E71T-1	1.2	DCEP	200	28	35	175	9.60	CO ₂	18	-	-

PAE Inspection By

Witness By

Approved By

SIGNATURE :

NAME :

DATE :



SIGNATURE :

NAME :

DATE :



SIGNATURE :

NAME :

DATE :

A

09/06/12

[illegible]



Testing Laboratory LPN Plate Mill Public Company Limited

199/9 Moo 4, Suksawad Rd., Pakkik ngbangplakod, Prasamutjedee, Samutprakarn 10290 Thailand.

Tel. 0 2815 6400-9 ext. 1179,1210. Direct Line and Fax. 0 2815 6415 , E-mail : qa@lpnrm.co.th



21716

Tension Testing Report

Customer : RENEWCOILS ENGINEERING & SUPPLY CO., LTD.
Address : 59 MOO 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000

Page 1 of 1
Report No. TS-12-06-108
Revision No. 0

Test Product	PQR No. RNC-PQR-FCW-001	Material Characterization	Plate
Project Name	N/A	Test Method	AWS D1.1/D1.1 M:2010
Project No.	N/A	Standard of Test	AWS D1.1/D1.1 M:2010
Material Specification	SS400	Type of Test	Reduced Section Tension
Dimension	25 mm Thk.	Equipment's Capacity	200 TONS
Date Received	19 June 2012	Equipment / Serial No.	SHIMADZU UH200A / 32098926
Date Tested	21 June 2012	Ambient Temperature	23 ± 5 °C

Item No.	Test No.	Specimen Dimension					Yield		Ultimate		Elongation (%)	Remark
		Diameter (mm)	Thickness (mm)	Width (mm)	Area (mm ²)	Gauge Length (mm)	Load (kN)	Strength (N/mm ²)	Load (kN)	Strength (N/mm ²)		
1	2369/12 TR1	N/A	24.84	20.37	505.991	N/A	N/A	N/A	244.381	482.976	N/A	OUT OF WELD
2	2369/12 TR2	N/A	24.85	20.52	509.922	N/A	N/A	N/A	244.381	479.253	N/A	OUT OF WELD

The unit of N/mm² is equivalent to MPa

Additional details : WPS No. RNC-WPS-FCW-001

Welding Process / Position : FCAW / 2G

Filler Metal/Electrode Spec't : E71T-1/DW100

Reported by :

(Mr.Keerati Sa-ngoen)

Laboratory Section Manager

Date 29 / 06 / 12

Approved by :

(Mr.Pavaret Preedawiphat)

Deputy Managing Director

Date 29 / 06 / 12

LPN PLATE MILL

- Notes**
1. The above results are valid exclusively for tested samples as mentioned in this report.
 2. Partial publicity of the results on testing is prohibited without the written permission from LPN LABORATORY
 3. This report shall not be reproduced in whole or in part without the written approval of LPN LABORATORY.



Testing Laboratory LPN Plate Mill Public Company Limited

199/9 Moo 4, Suksawad Rd., Pakklongbangplakod, Prasamutjedee, Samutprakarn 10290 Thailand.

Tel. 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax. 0 2815 6415, E-mail : qa@lpnrm.co.th



21724

Bend Testing Report

Customer : RENEWCOILS ENGINEERING & SUPPLY CO., LTD.

Page 1 of 1

Address : 59 MOO 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000

Report No. BD-12-06-106

Revision No. 0

Test Product	PQR No. RNC-PQR-FCW-001	Material Characterization	Plate
Project Name	N/A	Test Method	AWS D1.1/D1.1 M:2010
Project No.	N/A	Standard of Test	AWS D1.1/D1.1 M:2010
Material Specification	SS400	Type of Jig	Guided-Bend Roller Jig
Dimension	25 mm Thk.	Diameter of Mandrel	38.1 mm
Date Received	19 June 2012	Bend Angle	180 Degree.
Date Tested	21 June 2012	Equipment / Serial No.	SHIMADZU UH200A/32098926

Item No.	Test No.	Type of Bend	Size of Specimen (ThickxWidthxLength)	Indication	Location of Indication	Remark
1	2369/12 BS1	Transverse side bend	10.14x23.48x162.42 mm	No Open Discontinuity	N/A	-
2	2369/12 BS2	Transverse side bend	10.03x23.45x162.90 mm	No Open Discontinuity	N/A	-
3	2369/12 BS3	Transverse side bend	10.03x23.32x162.35 mm	No Open Discontinuity	N/A	-
4	2369/12 BS4	Transverse side bend	10.12x23.30x161.67 mm	Open Discontinuity 1.10 mm	WELD ZONE	-

Additional details : WPS No. RNC-WPS-FCW-001

Welding Process / Position : FCAW / 2G

Filler Metal/Electrode Spec't : E71T-1/DW100

Reported by : 

(Mr.Keerati Sa-ngoen)

Laboratory Section Manager

Date 29 / 06 / 12



Approved by : 

(Mr.Pavaret Preedawiphat)

Deputy Managing Director

Date 29 / 06 / 12

LPN PLATE MILL

Notes 1 The above results are valid exclusively for tested samples as mentioned in this report.

2 Partial publicity of the results on testing is prohibited without the written permission from LPN LABORATORY

3 This report shall not be reproduced in whole or in part without the written approval of LPN LABORATORY



Testing Laboratory LPN Plate Mill Public Company Limited

199/9 Moo 4, Suksawad Rd., Pakklongbangplakod, Prasamutjedee, Samutprakarn 10290 Thailand.

Tel. 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax. 0 2815 6415, E-mail : qa@lpnmpm.co.th



21726

Impact Testing Report

Customer : RENEWCOILS ENGINEERING & SUPPLY CO., LTD.

Page 1 of 1

Address : 59 MOO 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000

Report No. IM-12-06-134

Revision No. 0

Test Product	PQR No. RNC-PQR-FCW-001	Material Characterization	Plate
Project Name	N/A	Test Method	ASTM A370-10 and ASTM E23M-07a ^{sl}
Project No.	N/A	Standard of Test	AWS D1.1/D1.1 M:2010
Material Specification	SS400	Type of Test	Charpy V-Notch Impact
Dimension	25 mm Thk.	Equipment / Serial No.	Tinius Olsen / 221811
Date Received	19 June 2012	Equipment 's Capacity	542 Joules
Date Tested	21 June 2012	Ambient Temperature	23 ± 5 °C

Item No.	Test No.	Test Temperature (°C)	Size of Specimen			Results		Location of Test	Remark
			Thickness (mm)	Width (mm)	Length (mm)	Energy Absorbed (Joules)	Average (Joules)		
1	2369/12-1	-20°C	10.013	10.017	54.75	73.92	53.52	Weld Metal	-
2	2369/12-2	-20°C	10.017	10.019	54.81	44.63			
3	2369/12-3	-20°C	10.014	10.013	54.84	42.00			
4	2369/12-1	-20°C	10.010	10.014	54.81	60.30	69.16	Fusion Line+1mm	-
5	2369/12-2	-20°C	10.008	10.016	54.84	71.46			
6	2369/12-3	-20°C	10.015	10.015	54.94	75.72			
7	2369/12-1	-20°C	10.011	10.014	54.77	26.51	23.11	Fusion Line+5mm	-
8	2369/12-2	-20°C	10.011	10.018	54.91	25.01			
9	2369/12-3	-20°C	10.012	10.017	54.86	17.80			

Additional details : WPS No. RNC-WPS-FCW-001

Welding Process / Position : FCAW / 2G

Filler Metal/Electrode Spec't : E71T-1/DW100

Reported by :

(Mr.Keerati Sa-ngoen)

Laboratory Section Manager

Date 29 / 06 / 12



Approved by :

(Mr.Pavaret Freedawiphat)

Deputy Managing Director

Date 29 / 06 / 12

LPN PLATE MILL

Notes 1 The above results are valid exclusively for tested samples as mentioned in this report.

2. Partial publicity of the results on testing is prohibited without the written permission from LPN LABORATORY.

3. This report shall not be reproduced in whole or in part without the written approval of LPN LABORATORY.



Testing Laboratory LPN Plate Mill Public Company Limited

199/9 Moo 4, Suksawad Rd., Pakklongbangplakod, Prasamutjedee, Samutprakarn 10290 Thailand.

Tel. 0 2815 6400-9 ext. 1179, 1210, Direct Line Fax. 0 2815 6415 , E-mail : qa@lpnpm.co.th

11068

Macrostructure Analysis Report

Page 1 of 1

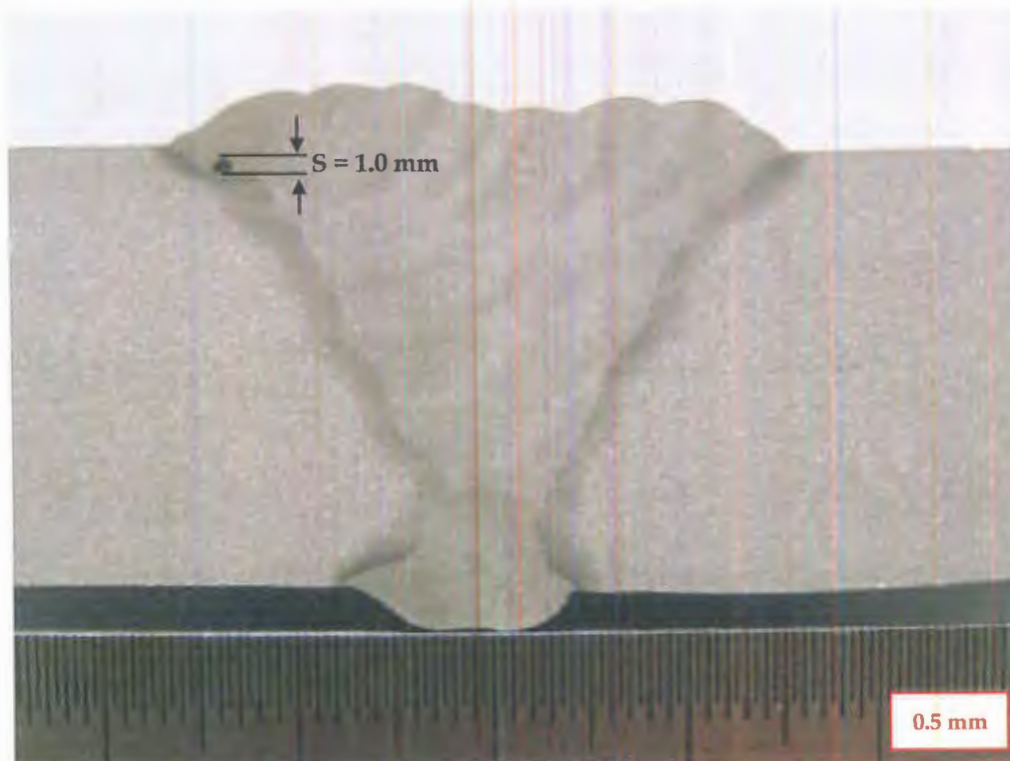
Customer : RENEWCOILS ENGINEERING & SUPPLY CO.,LTD.

Report No. MA-12-06-105

Address : 59 Moo 8 HIGHWAY 36 T.TABMA A.MUANG,RAYONG 21000 Thailand.

Revision No. 0

Test Product	PQR No. RNC-PQR-FCW-001	Material Characterization	Plate
Test No.	2369/12 MA	Test Method	AWS D1.1/D1.1 M : 2010
Project Name	N/A	Standard of Test	AWS D1.1/D1.1 M : 2010
Project No.	N/A	Etchant Reagent	75ml H ₂ O+25ml HNO ₃ (65%)
Material Specification	SS400	Temperature	28 °C
Dimension	25 mm. Thk.	Time of Etching	5 minutes
Welding Process/Position	FCAW/2G	Surface Preparation	Polishing with sand paper
Type of Joint	Butt Joint		240, 400, 600, 800, 1000
Date Received	19 June 2012	Date Tested	21 June 2012



Additional details : The macrostructure shows S = Slag inclusion.
Filler/Electrode Spec't : E71T-1 DW100

Reported by :
(Mr.Keerati Sa-ngoen)
Laboratory Section Manager
Date : 29/06/12



Approved by :
(Mr.Pavaret Preedawiphat)
Deputy Managing Director
Date : 29/06/12

LPN PLATE MILL

Notes

1. The above results are valid exclusively for tested samples as mentioned in this report.
2. Partial publicity of the results on testing is prohibited without the written permission from LPN LABORATORY.
3. This report shall not be reproduced in whole or in part without the written approval of LPN LABORATORY.



Testing Laboratory LPN Plate Mill Public Company Limited 12410

199/9 Moo 4, Suksawad Rd., Pakklongbangplakod, Prasamutjedee, Samutprakarn 10290 Thailand.

Tel. 0 2815 6400-9 ext. 1179, 1210, Direct Line Fax. 0 2815 6415 , E-mail : qa@lpnpm.co.th

Hardness Testing Report

Page 1 of 1

Customer : RENEWCOILS ENGINEERING & SUPPLY CO.,LTD.

Report No. HN-12-06-083

Address : 59 Moo 8 HIGHWAY 36 T.TABMA A.MUANG,RAYONG 21000 Thailand.

Revision No. 0

Test Product	PQR No. RNC-PQR-001	Material Characterization	Plate
Test No.	2369/12 HN	Test Method	ASTM E92-82 : 2003
Project Name	N/A	Standard of Test	ASTM Sec. IX : 2010
Project No.	N/A	Type of test	Vicker Hardness Test
Material Specification	SS400	Equipment	Mitutoyo, HV-115
Dimension	25 mm. Thk.	Indenter Size	Diamond, pyramid 136 deg.
Welding Process/Position	FCAW/2G	Test Load	HV 10 Kgf.
Type of Joint	Butt Joint	Total Force Dwell Time	15 s
Date Received	19 June 2012	Cal. Block	No. 62137 No. A62072
Date Tested	21 June 2012	Ambient Temperature	25.8 °C

Test Location

Total = 30 points



Point No.	Location	Hardness value Value (Scale HV)	Point No.	Location	Hardness value Value (Scale HV)	Point No.	Location	Hardness value Value (Scale)
1	BASE	131.7	16	BASE	132.1	*****		
2	BASE	131.0	17	BASE	135.0			
3	BASE	136.6	18	BASE	132.4			
4	HAZ	158.5	19	HAZ	158.4			
5	HAZ	171.7	20	HAZ	177.0			
6	HAZ	193.0	21	HAZ	214.8			
7	WELD	196.7	22	WELD	223.4			
8	WELD	200.0	23	WELD	223.9			
9	WELD	198.4	24	WELD	223.6			
10	HAZ	190.8	25	HAZ	191.4			
11	HAZ	183.6	26	HAZ	175.5			
12	HAZ	173.6	27	HAZ	165.5			
13	BASE	130.3	28	BASE	133.3			
14	BASE	129.5	29	BASE	133.2			
15	BASE	129.2	30	BASE	132.2			

Additional details : Filler/Electrode Spec't : E71T-1 DW100

Reported by :
(Mr.Keerati Sa-ngoen)
Laboratory Section Manager
Date : 29/06/12



Approved by :
(Mr.Pavaret Preedawiphat)
Deputy Managing Director
Date : 29/06/12

Notes

1. The above results are valid exclusively for tested samples as mentioned in this report.
2. Partial publicity of the results on testing is prohibited without the written permission from LPN LABORATORY.
3. This report shall not be reproduced in whole or in part without the written approval of LPN LABORATORY.

LPN PLATE MILL

WELDING PROCEDURE SPECIFICATION (WPS) Yes ☐
PREQUALIFIED _____ QUALIFIED BY TESTING _____
or PROCEDURE QUALIFICATION RECORD (PQR) Yes ☒
(ACCORDANCE WITH THE REQUIREMENT OF SEC'T 4 AWS D1.1 2008)

Identification # RNC-WPS-FCAW-001
Revision 1 Date 9 Jun 12 By RNC
QA/QC Date 9 Jun 12
Type ☐ Manual ☒ Semi-Automatic
☐ Machine ☐ Automatic

Company Name RENEWCOILS ENGINEERING & SUPPLY CO.,LTD.
Welding Process(es) FLUX CORED ARC WELDING (FCAW)
Supporting PQR No.(s) RNC-PQR-FCAW-001

JOINT DESIGN USED

Type : Single V-Groove Butt Weld
☒ Single ☐ Double weld
Backing : ☐ Yes ☒ No
Backing Material : None
Root Opening : 3.0 mm. Root Face Dimension : 3.0 mm.
Groove Angle : 60° ± 5 Radius (J-U) -
Back Gouging ☒ Yes ☐ No
Method : Grinding

BASE METALS

Material Spec. Carbon Steel
Type or Grade SS 400
Thickness : Groove 25 mm. Fillet : -
Diameter (Pipe) -

FILLER METALS

AWS Specification A 5.20
AWS Classification E71T-1

SHIELDING

Flux NONE Gas CO₂
Composition 99.99 %
Electrode-Flux(Class) - Flow Rate 18 L/min.
Gas Cup Size N/A

PREHEAT

Preheat Temp.: Ambient
Interpass Temp., Min 118°C Max 194°C

POSITION

Position of Groove 3G Fillet V
Vertical Progression ☒ Up ☐ Down

ELECTRICAL CHARACTERISTICS

Transfer Mode (GMAW) ☐ Short-Circuiting
☐ Globular ☐ Spray/Pulse
Current ☐ AC ☐ DCEP ☒ DCEN ☐ Pulsed
Other -
Tungsten Electrode (GTAW)
Size None
Type None

TECHNIQUE

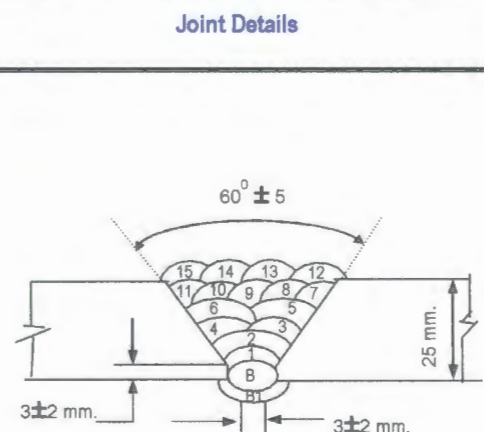
Stringer or Weave Bead : Stringer
Multi-pass or Single Pass (per side) Multipass
Number of Electrodes Single
Electrode Spacing Longitudinal N/A
Lateral N/A
Angle N/A
Contact End to Work Distance 15 mm.
Peening None
Interpass Cleaning : Slags remove by Grinding / Wire Brush

POSTWELD HEAT TREATMENT

Temperature N/A
Time N/A

WELDING PROCEDURE

Pass or Weld Layer(s)	Process	Filler Metals Class	Dia. (mm.)	Current Type & Polarity	Amperage or Wire speed	Volts	Travel Speed (cm / min)	Heat Input (kj/cm)
1	FCAW	E71T-1	1.2	DCEP	140	22	18	10.37
2	FCAW	E71T-1	1.2	DCEP	200	30	17	21.18
3	FCAW	E71T-1	1.2	DCEP	210	31	30	13.02
4	FCAW	E71T-1	1.2	DCEP	210	30	30	12.60
5	FCAW	E71T-1	1.2	DCEP	210	30	28	13.50
6	FCAW	E71T-1	1.2	DCEP	220	31	30	13.64
7	FCAW	E71T-1	1.2	DCEP	220	30	30	13.20
8	FCAW	E71T-1	1.2	DCEP	220	30	30	13.20
9	FCAW	E71T-1	1.2	DCEP	220	31	35	11.69
10	FCAW	E71T-1	1.2	DCEP	200	29	25	13.92
11	FCAW	E71T-1	1.2	DCEP	180	29	35	8.95
12	FCAW	E71T-1	1.2	DCEP	190	26	20	14.82
13	FCAW	E71T-1	1.2	DCEP	190	26	22	13.47
14	FCAW	E71T-1	1.2	DCEP	190	26	23	12.89
15	FCAW	E71T-1	1.2	DCEP	190	26	23	12.83
Back Weld	FCAW	E71T-1	1.2	DCEP	190	26	25	11.86
B1	FCAW	E71T-1	1.2	DCEP	190	26	25	11.86





PAE TECHNICAL SERVICE COMPANY LIMITED
69 Soi On-nuch 64, Srinakarin Road, Suanluang, Suanluang, Bangkok 10250
Tel. : (662) 322-0222 (Auto 20 lines) Fax. : (662) 721-2577

PQR No. : RNC-PQR-001

PROCEDURE QUALIFICATION RECORD (PQR)

NDT and Mechanical Test Result

TENSILE TEST (Report No : TS-12-06-109)						
Specimen No.	Width mm.	Thickness mm.	Area	Ultimate Load (KN)	Ultimate Strength (N/mm. ²)	Character of Failure and Location
2370/12TR1	20.26	24.46	495.560	238.301	480.873	OUT OF WELD
2370/12TR2	20.36	24.69	502.688	241.047	479.516	OUT OF WELD

GUIDE BEND TEST (Report No : BD-12-06-107)			
Specimen No	Type of Bend	Test Result	Location of Indication
2370/12BS1	Transverse Side Bend	No Open Discontinuity	N/A
2370/12BS2	Transverse Side Bend	Open Discontinuity 1.2,1.2 mm	WELD ZONE
2370/12BS3	Transverse Side Bend	Open Discontinuity 1.49 mm	WELD ZONE
2370/12BS4	Transverse Side Bend	Open Discontinuity 1.6 mm	WELD ZONE

Visual Inspection, PQR Report No:

PQR-008

Appearance: Acceptable
Undercut/Slags: No Indication
Surface Porosity: No Indication
Test Date: 15 Jun 12
Witness By: Mr.Natthawut M.
Organization: PAE Technical Service Co.,Ltd.

Radiographic Examination

RT Report No: PQR-008 Result: Accepted
Organization: PAE Technical Service Co.,Ltd.

Other Tests

1) Vicker Hardness Test

Report No: HN-12-06-084

2) Macrostructure Examination

Report No: MA-12-06-106

3) Fillet Weld Test

Report No: N.A.

4) Impact Test

Report No: IM-12-06-135

Welder's Name: Mr.Prasong P.

Welder No: -

Mechanical Test Conducted By: LPN Plate Mill Public Company Limited

Test Conducted By: Mr.Keerati Sa-ngoen

Report No: -

We, the undersigned, certify the statements in this record are correct and that the test welds were prepared, welded, and Tested in conformance with the requirements of Clause Section 4 of AWS D1.1 Structural Welding Code-Steel,

Manufacturer by
Rayong Maintenance & Contracting Co.,Ltd.

Signature:

Name

Date



Contractor by
PAE Technical Service Co.,Ltd.

Signature:

Name

Date





PAE TECHNICAL SERVICE CO.,LTD

NDT, TESTING, INSPECTION & ENGINEERING CONSULTANT

69 SOI.OONNUT 64 (SUKSAMARN) SRINAKARIN RD, SUANLUANG,BANGKOK 10250

Tel : 02) 322-0222, Fax : 02)721-2577, RAYONG (038) 607-402-3

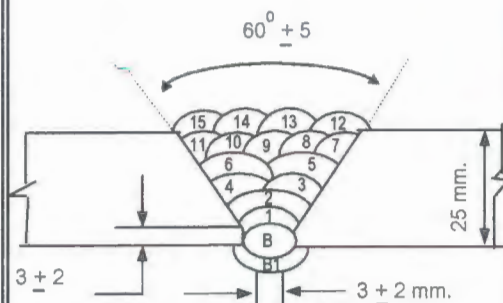
WELDER PROCEDURE QUALIFICATION RECORD

Page No. 1 of 2

Client : Renew Coils Engineering & Supply Co.,Ltd.	Project Name : PQR	Location : RNC Work Shop	Report No. PQR-008
Customer Order No. : N/A	WPS No. : RNC-WPS-FCAW-001	Date : 15-Jun-12	

Welder No.	Welder Name	Welding Process	Test Position	Applicable Standard	Material		Electrode Specification	Results	
					Spec	Size		Visual	RT
-	MR.PRASONG P.	FCAW	3G	AWS D1.1	SS400	350x150x25	E71T-1	ACCEPTED	ACCEPTED

Joint Details



Layer (S)	Process	Filler Metal		Current		Volt	Travel Speed cm/min	Interpas Temp °C	Heat Input (KJ / cm)	Shielding		Backing	
		Class	Dia. (mm.)	Type Polarity	Amp					GAS	Flow Rate (l/min)	GAS	Flow Rate (l/min)
1	FCAW	E71T-1	1.2	DCEP	140	22	18	118	10.37	CO ₂	18	-	-
2	FCAW	E71T-1	1.2	DCEP	200	30	17	140	21.18	CO ₂	18	-	-
3	FCAW	E71T-1	1.2	DCEP	210	31	30	177	13.02	CO ₂	18	-	-
4	FCAW	E71T-1	1.2	DCEP	210	30	30	178	12.60	CO ₂	18	-	-
5	FCAW	E71T-1	1.2	DCEP	210	30	28	191	13.50	CO ₂	18	-	-
6	FCAW	E71T-1	1.2	DCEP	220	31	30	194	13.64	CO ₂	18	-	-
7	FCAW	E71T-1	1.2	DCEP	220	30	30	174	13.20	CO ₂	18	-	-
8	FCAW	E71T-1	1.2	DCEP	220	30	30	179	13.20	CO ₂	18	-	-
9	FCAW	E71T-1	1.2	DCEP	220	31	35	192	11.69	CO ₂	18	-	-
10	FCAW	E71T-1	1.2	DCEP	200	29	25	174	13.92	CO ₂	18	-	-

PAE Inspection By

Witness By

Approved By

SIGNATURE :

NAME :

DATE :



SIGNATURE :

NAME :

DATE :



SIGNATURE :

NAME :

DATE :

[Signature]

09/06/12



PAE TECHNICAL SERVICE CO.,LTD

NDT, TESTING, INSPECTION & ENGINEERING CONSULTANT

69 SOI.OONNUT 64 (SUKSAMARN) SRINAKARIN RD, SUANLUANG,BANGKOK 10250

Tel : 02) 322-0222, Fax : 02)721-2577, RAYONG (038) 607-402-3

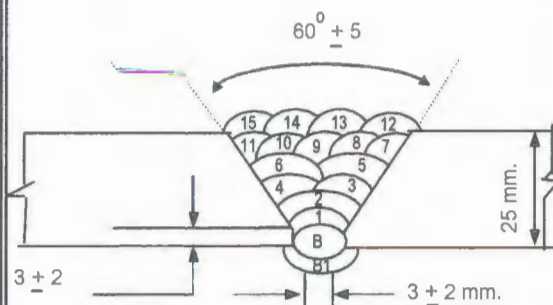
WELDER PROCEDURE QUALIFICATION RECORD

Page No. 2 of 2

Client : Renew Coils Engineering & Supply Co.,Ltd.	Project Name : PQR	Location : RNC Work Shop	Report No. PQR-008
Customer Order No. : N/A	WPS No. : RNC-WPS-FCAW-001	Date : 15-Jun-12	

Welder No.	Welder Name	Welding Process	Test Position	Applicable Standard	Material		Electrode Specification	Results	
					Spec	Size		Visual	RT
-	MR.PRASONG P.	FCAW	3G	AWS D1.1	SS400	350x150x25	E71T-1	ACCEPTED	ACCEPTED

Joint Details



Layer (S)	Process	Filler Metal		Current		Volt	Travel Speed cm/min	Interpas Temp °C	Heat Input (KJ / cm)	Shielding		Backing	
		Class	Dia. (mm.)	Type Polarity	Amp					GAS	Flow Rate (l/min)	GAS	Flow Rate (l/min)
11	FCAW	E71T-1	1.2	DCEP	180	29	35	184	8.95	CO ₂	18	-	-
12	FCAW	E71T-1	1.2	DCEP	190	26	20	174	14.82	CO ₂	18	-	-
13	FCAW	E71T-1	1.2	DCEP	190	26	22	180	13.47	CO ₂	18	-	-
14	FCAW	E71T-1	1.2	DCEP	190	26	23	191	12.89	CO ₂	18	-	-
15	FCAW	E71T-1	1.2	DCEP	190	26	23	190	12.83	CO ₂	18	-	-
B	FCAW	E71T-1	1.2	DCEP	190	26	25	153	11.86	CO ₂	18	-	-
B1	FCAW	E71T-1	1.2	DCEP	190	26	25	140	11.86	CO ₂	18	-	-

PAE Inspection By

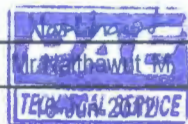
Witness By

Approved By

SIGNATURE :

NAME :

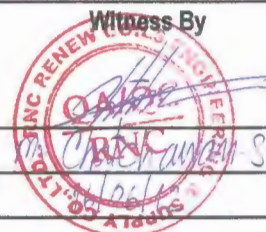
DATE :



SIGNATURE :

NAME :

DATE :



SIGNATURE :

NAME :

DATE :

[Signature]

09/06/12

PAE TECHNICAL SERVICE CO.,LTD.

NDT, TESTING, INSPECTION & ENGINEERING CONSULTANT

TEL : 02) 322-0222 FAX : 02) 721-2577 RAYONG TEL : 038) 607402-3 Fax : 038) 607402 SRIRACHA : 038) 330220 330229

Page : 1 Of 1

[illegible]



Testing Laboratory LPN Plate Mill Public Company Limited

199/9 Moo 4, Suksawad Rd., Pakklongbangplakod, Prasamutjedee, Samutprakarn 10290 Thailand.

Tel. 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax. 0 2815 6415 , E-mail : qa@lpnrm.co.th



21717

Tension Testing Report

Customer : RENEWCOILS ENGINEERING & SUPPLY CO., LTD.
Address : 59 MOO 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000

Page 1 of 1
Report No. TS-12-06-109
Revision No. 0

Test Product	PQR No. RNC-PQR-FCW-001	Material Characterization	Plate
Project Name	N/A	Test Method	AWS D1.1/D1.1 M:2010
Project No.	N/A	Standard of Test	AWS D1.1/D1.1 M:2010
Material Specification	SS400	Type of Test	Reduced Section Tension
Dimension	25 mm Thk.	Equipment's Capacity	200 TONS
Date Received	19 June 2012	Equipment / Serial No.	SHIMADZU UH200A / 32098926
Date Tested	21 June 2012	Ambient Temperature	23 ± 5 °C

Item No.	Test No.	Specimen Dimension					Yield		Ultimate		Elongation (%)	Remark
		Diameter (mm)	Thickness (mm)	Width (mm)	Area (mm ²)	Gauge Length (mm)	Load (kN)	Strength (N/mm ²)	Load (kN)	Strength (N/mm ²)		
1	2370/12 TR1	N/A	24.46	20.26	495.560	N/A	N/A	N/A	238.301	480.873	N/A	OUT OF WELD
2	2370/12 TR2	N/A	24.69	20.36	502.688	N/A	N/A	N/A	241.047	479.516	N/A	OUT OF WELD

The unit of N/mm² is equivalent to MPa

Additional details : WPS No. RNC-WPS-FCW-001

Welding Process / Position : FCAW / 3G

Filler Metal/Electrode Spec't : E71T-1/DW100

Reported by :

(Mr.Keerati Sa-ngoen)

Laboratory Section Manager

Date 29 / 06 / 12



Approved by :

(Mr.Pavaret Preedawiphat)

Deputy Managing Director

Date 29 / 06 / 12

LPN PLATE MILL

Notes 1. The above results are valid exclusively for tested samples as mentioned in this report.

2. Partial publicity of the results on testing is prohibited without the written permission from LPN LABORATORY.

3. This report shall not be reproduced in whole or in part without the written approval of LPN LABORATORY.



Testing Laboratory LPN Plate Mill Public Company Limited

199/9 Moo 4, Suksawad Rd., Pakklongbangplakod, Prasamutjedee, Samutprakarn 10290 Thailand.

Tel. 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax. 0 2815 6415, E-mail : qa@lpnppn.co.th



21725

Bend Testing Report

Customer : RENEWCOILS ENGINEERING & SUPPLY CO., LTD.

Page 1 of 1

Address : 59 MOO 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000

Report No. BD-12-06-107

Revision No. 0

Test Product	PQR No. RNC-PQR-FCW-001	Material Characterization	Plate
Project Name	N/A	Test Method	AWS D1.1/D1.1 M:2010
Project No.	N/A	Standard of Test	AWS D1.1/D1.1 M:2010
Material Specification	SS400	Type of Jig	Guided-Bend Roller Jig
Dimension	25 mm Thk.	Diameter of Mandrel	38.1 mm
Date Received	19 June 2012	Bend Angle	180 Degree.
Date Tested	21 June 2012	Equipment / Serial No.	SHIMADZU UH200A/32098926

Item No.	Test No.	Type of Bend	Size of Specimen (ThickxWidthxLength)	Indication	Location of Indication	Remark
1	2370/12 BS1	Transverse side bend	10.00x23.50x161.51 mm	No Open Discontinuity	N/A	
2	2370/12 BS2	Transverse side bend	10.12x23.45x162.36 mm	Open Discontinuity	WELD ZONE	1.2, 1.2 mm
3	2370/12 BS3	Transverse side bend	10.11x23.47x162.30 mm	Open Discontinuity	WELD ZONE	1.49 mm
4	2370/12 BS4	Transverse side bend	10.03x23.48x161.67 mm	Open Discontinuity	WELD ZONE	1.6 mm

Additional details : WPS No. RNC-WPS-FCW-001

Welding Process / Position : FCAW / 3G

Filler Metal/Electrode Spec't : E71T-1/DW100

Reported by :

(Mr.Keerati Sa-ngoen)

Laboratory Section Manager

Date 29 / 06 / 12



Approved by :

(Mr.Pavaret Freedawiphat)

Deputy Managing Director

Date 29 / 06 / 12

LPN PLATE MILL

Notes 1 The above results are valid exclusively for tested samples as mentioned in this report.

2 Partial publicity of the results on testing is prohibited without the written permission from LPN LABORATORY.

3 This report shall not be reproduced in whole or in part without the written approval of LPN LABORATORY.



Testing Laboratory LPN Plate Mill Public Company Limited

199/9 Moo 4, Suksawad Rd., Pakklongbangplakod, Prasamutjedee, Samutprakarn 10290 Thailand.

Tel. 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax. 0 2815 6415, E-mail : qa@lpnrm.co.th



Impact Testing Report

Customer : RENEWCOILS ENGINEERING & SUPPLY CO., LTD.

Page 1 of 1

Address : 59 MOO 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000

Report No. IM-12-06-135

Revision No. 0

Test Product	PQR No. RNC-PQR-FCW-001	Material Characterization	Plate
Project Name	N/A	Test Method	ASTM A370-10 and ASTM E23M-07a ¹
Project No.	N/A	Standard of Test	AWS D1.1/D1.1 M:2010
Material Specification	SS400	Type of Test	Charpy V-Notch Impact
Dimension	25 mm Thk.	Equipment / Serial No.	Tinius Olsen / 221811
Date Received	19 June 2012	Equipment 's Capacity	542 Joules
Date Tested	21 June 2012	Ambient Temperature	23 ± 5 °C

Item No.	Test No.	Test Temperature (°C)	Size of Specimen			Results		Location of Test	Remark
			Thickness (mm)	Width (mm)	Length (mm)	Energy Absorbed (Joules)	Average (Joules)		
1	2370/12-1	-20°C	10.014	10.014	54.89	63.37	63.31	Weld Metal	-
2	2370/12-2	-20°C	10.014	10.015	54.84	67.46			
3	2370/12-3	-20°C	10.014	10.014	54.88	59.10			
4	2370/12-1	-20°C	10.014	10.016	54.88	59.10	59.99	Fusion Line+1mm	-
5	2370/12-2	-20°C	10.012	10.016	54.91	55.41			
6	2370/12-3	-20°C	10.013	10.015	54.96	65.47			
7	2370/12-1	-20°C	10.009	10.014	54.81	31.06	28.54	Fusion Line+5mm	-
8	2370/12-2	-20°C	10.012	10.016	54.86	23.21			
9	2370/12-3	-20°C	10.012	10.013	54.82	31.36			

Additional details : WPS No. RNC-WPS-FCW-001

Welding Process / Position : FCAW / 3G

Filler Metal/Electrode Spec't : E71T-1/DW100

Reported by : 

(Mr.Keerati Sa-ngoen)

Laboratory Section Manager

Date 29 / 06 / 12



Approved by : 

(Mr.Pavaree Freedawiphat)

Deputy Managing Director

Date 29 / 06 / 12

LPN PLATE MILL

Notes 1 The above results are valid exclusively for tested samples as mentioned in this report.

2. Partial publicity of the results on testing is prohibited without the written permission from LPN LABORATORY.

3. This report shall not be reproduced in whole or in part without the written approval of LPN LABORATORY.



LPN PLATE MILL

Testing Laboratory LPN Plate Mill Public Company Limited

199/9 Moo 4, Suksawad Rd., Pakklongbangplakod, Prasamutjedee, Samutprakarn 10290 Thailand.

Tel. 0 2815 6400-9 ext. 1179, 1210, Direct Line Fax. 0 2815 6415 , E-mail : qa@lpnpm.co.th

11069

Macrostructure Analysis Report

Page 1 of 1

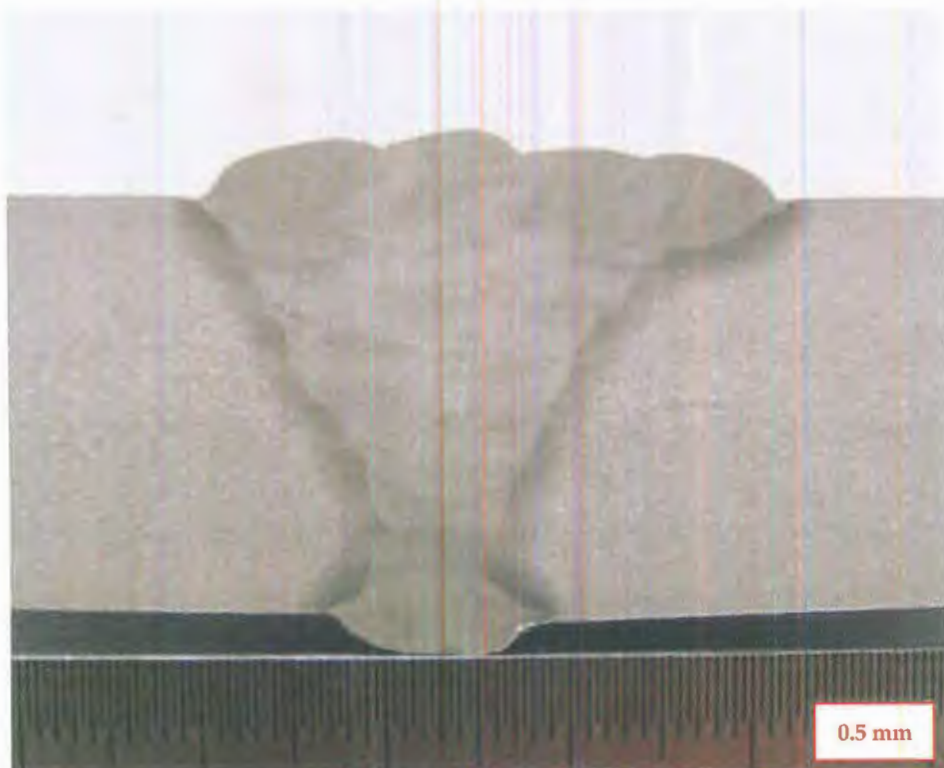
Customer : RENEWCOILS ENGINEERING & SUPPLY CO.,LTD.

Report No. MA-12-06-106

Address : 59 Moo 8 HIGHWAY 36 T.TABMA A.MUANG,RAYONG 21000 Thailand.

Revision No. 0

Test Product	PQR No. RNC-PQR-FCW-001	Material Characterization	Plate
Test No.	2370/12 MA	Test Method	AWSD1.1/D1.1M : 2010
Project Name	N/A	Standard of Test	AWSD1.1/D1.1M : 2010
Project No.	N/A	Etchant Reagent	75ml H ₂ O+25ml HNO ₃ (65%)
Material Specification	SS400	Temperature	28 °C
Dimension	25 mm. Thk.	Time of Etching	5 minutes
Welding Process/Position	FCAW/3G	Surface Preparation	Polishing with sand paper
Type of Joint	Butt Joint		240, 400, 600, 800, 1000
Date Received	19 June 2012	Date Tested	21 June 2012



Additional details : The macrostructure shows the complete fusion.Weld metal and heat affected zone were free of cracks
Filler/Electrode Spec't : E71T-1 DW100

Reported by :

(Mr.Keerati Sa-ngoen)

Laboratory Section Manager

Date : 29/06/12



Approved by :

(Mr.Pavaret Freedawiphat)

Deputy Managing Director

Date : 29/06/12

LPN PLATE MILL**Notes**

1. The above results are valid exclusively for tested samples as mentioned in this report.
2. Partial publicity of the results on testing is prohibited without the written permission from LPN LABORATORY.
3. This report shall not be reproduced in whole or in part without the written approval of LPN LABORATORY.



Testing Laboratory LPN Plate Mill Public Company Limited 12411

199/9 Moo 4, Suksawad Rd., Pakklongbangplakod, Prasamutjedee, Samutprakarn 10290 Thailand.

Tel. 0 2815 6400-9 ext. 1179, 1210, Direct Line Fax. 0 2815 6415 , E-mail : qa@lpnpm.co.th

Hardness Testing Report

Page 1 of 1

Customer : RENEWCOILS ENGINEERING & SUPPLY CO.,LTD.

Report No. HN-12-06-084

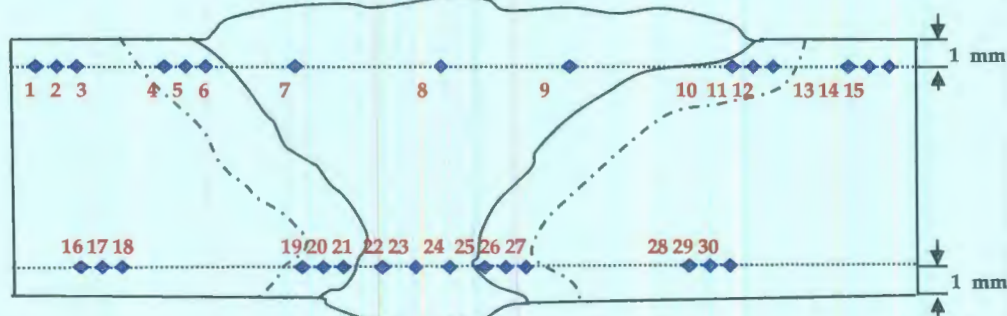
Address : 59 Moo 8 HIGHWAY 36 T.TABMA A.MUANG,RAYONG 21000 Thailand.

Revision No. 0

Test Product	PQR No. RNC-PQR-FCW-001	Material Characterization	Plate
Test No.	2370/12 HN	Test Method	ASTM E92-82 : 2003
Project Name	N/A	Standard of Test	ASTM Sec. IX : 2010
Project No.	N/A	Type of test	Vicker Hardness Test
Material Specification	SS400	Equipment	Mitutoyo, HV-115
Dimension	25 mm. Thk.	Indenter Size	Diamond, pyramid 136 deg.
Welding Process/Position	FCAW/3G	Test Load	HV 10 Kgf.
Type of Joint	Butt Joint	Total Force Dwell Time	15 s
Date Received	19 June 2012	Cal. Block	No. 62137 No. A62072
Date Tested	21 June 2012	Ambient Temperature	25.8 °C

Test Location

Total = 30 points



Point No.	Location	Hardness value Value (Scale HV)	Point No.	Location	Hardness value Value (Scale HV)	Point No.	Location	Hardness value Value (Scale)
1	BASE	137.3	16	BASE	130.5	*****		
2	BASE	136.6	17	BASE	130.0			
3	BASE	136.6	18	BASE	131.6			
4	HAZ	168.4	19	HAZ	163.0			
5	HAZ	183.5	20	HAZ	177.1			
6	HAZ	204.4	21	HAZ	200.1			
7	WELD	201.2	22	WELD	215.7			
8	WELD	180.3	23	WELD	216.8			
9	WELD	209.6	24	WELD	212.9			
10	HAZ	184.1	25	HAZ	190.8			
11	HAZ	170.8	26	HAZ	157.3			
12	HAZ	164.9	27	HAZ	150.1			
13	BASE	130.6	28	BASE	132.5			
14	BASE	131.1	29	BASE	133.7			
15	BASE	132.3	30	BASE	132.3			

Additional details : Filler/Electrode Spec't : E71T-1 DW100

Reported by :

(Mr.Keerati Sa-ngoen)
Laboratory Section Manager
Date : 29/06/12



Approved by :

(Mr.Pavaret Preedawiphat)
Deputy Managing Director
Date : 29/06/12

Notes

1. The above results are valid exclusively for tested samples as mentioned in this report.
2. Partial publicity of the results on testing is prohibited without the written permission from LPN LABORATORY.
3. This report shall not be reproduced in whole or in part without the written approval of LPN LABORATORY.

LPN PLATE MILL



Inspection Certificate

检查证明书

Certificate No.

证明书编号: 11-273

Date of Issue

发行日: 2011.12.30

Trade Designation 品名	Dimension 尺寸(mm)	Lot No. 批号	Applicable Specification and Classification 适用标准及种类
DW-100	1.2	Q71Z0113331	AWS A5.20 E71T-1C

Chemical Composition
化学成分(%)

Elements 成分	C	Si	Mn	P	S	Cu	Ni	Cr	Mo	V				
Actual Result 实测值	0.07	0.52	1.33	0.016	0.013	0.03	0.01	0.03	<0.01	0.02				

Test Item 试验项目	Tensile Test of Deposited Metal 熔敷金属抗拉试验			Impact Test of Deposited Metal 熔敷金属冲击试验			Radiographic Test X射线检测	
Test Content 试验内容	Yield Strength 屈服强度(MPa)	Tensile Strength 抗拉强度(MPa)	Elongation 伸长率(%)	Test Temp. 试验温度(°C)	Impact Value 冲击功(J)			
Actual Result 实测值	498	575	25	-20	61	64 60 Avg. 62	—	

Welding Condition 焊接条件				Remarks 备注	
Type of Current 电流的种类	Amperage 焊接电流(A)	Arc Voltage 电弧电压(V)	Shielding Gas 保护气体		
DC-EP	300	34	CO2		

We hereby certify that the above test results are as described herein
and satisfy the requirement of the applicable specification.

焊丝试验结果如以上所记, 符合产品标准的要求, 特此证明。

KOBELCO WELDING OF QINGDAO CO., LTD.

青岛神钢焊接材料有限公司

Principal

检查负责人